

HAT-P-18b

R-2436

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-18_150803 · created 2026-08-22T01:25:05+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2457237.7007 +/- 0.00066 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.183 +/- 0.018
Transit depth (Rp/Rs)^2	3.37 +/- 0.65 [%]
Orbital Inclination (inc)	87.09 +/- 0.4
Airmass coefficient 1 (a1)	0.996 +/- 0.025
Airmass coefficient 2 (a2)	0.005 +/- 0.033
Scatter in the residuals of the lightcurve fit is	2.52 %
Best Comparison Star	#3 - [290, 88]
Optimal Aperture	2.94
Optimal Annulus	7.68
Transit Duration (day)	0.088 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.183 ± 0.018 vs 0.1356 (deviation 1.9σ)
Duration fit vs published	0.088 d vs 0.1129 d (deviation 1.63σ)
Mid-time O–C	0.4 min
Depth SNR	7.3
Residual scatter	2.52 %

Light curve

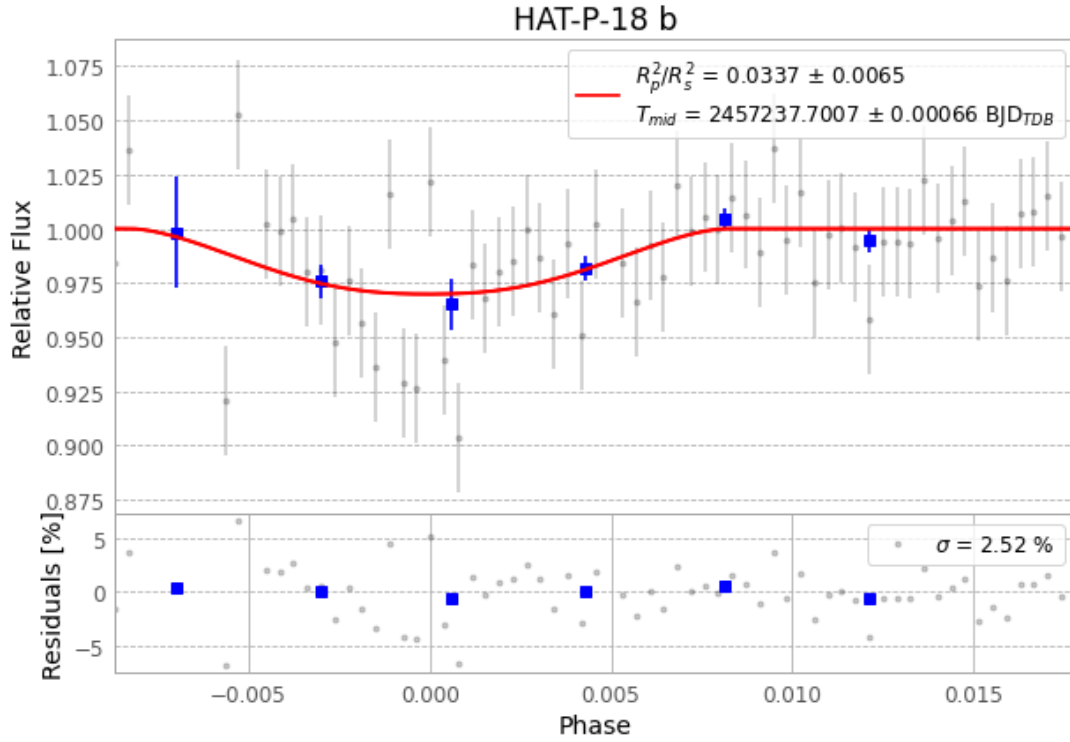


Figure 1 — FinalLightCurve_HAT-P-18 b_2015-08-02.png (SHA-256 2ae16d98fc98622c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [242, 213], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 4e0d6b3587f54609...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

71 FITS frames (47 MB), HATP-18150803033612.FITS → HATP-18150803070612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-18-150803.log (4f3645aabb1b...), FinalLightCurve_HAT-P-18 b_2015-08-02.png (2ae16d98fc98...), FinalLightCurve_HAT-P-18 b_2015-08-02.csv (a103db5f86bb...)

Compute resources

Wall time 133.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-22 01:25Z.